

Research

GIS-based assessment of economically feasible off-grid mini-grids in Ethiopia

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Abstract

Mini-grids (MGs) have emerged as a cost-effective solution for electrifying remote communities characterized by low population density and challenging geographical conditions. However, identifying optimal locations for sustainable MG deployment is complex due to technical, social, and economic factors. This study aims to identify economically feasible sites for renewable MGs in Ethiopia using the Open-Source Spatial Electrification Tool (OnSSET). This GIS-based framework systematically integrates key variables such as resource availability, population density, land cover, terrain slope, and proximity to existing infrastructure, using an equal weighting scheme across two grid proximity scenarios: 2.5 and 25 km from existing medium-voltage (MV) lines. To assess economic feasibility, the study calculated the Levelized Cost of Electricity (LCOE) for the selected sites. The results show that hydro MGs are the most cost-competitive option, with LCOE values between 0.09 and 0.16 \$/kWh. Under the 2.5 and 25 km grid proximity scenarios, 306 and 84 potential hydro MG sites are identified, capable of electrifying approximately 5 and 1.9 million people, respectively. Solar PV MGs also demonstrate significant potential, with LCOE values ranging from 0.15–0.22 \$/kWh. Solar MGs could electrify 7.2 million people under the 2.5 km scenario and 3.2 million under the 25 km scenario. Wind MG, with LCOE values ranging from 0.12–1.75 \$/kWh, could provide electricity to 4.8 million people under the 2.5 km scenario and 3.1 million under the 25 km scenario. This study provides a pre-feasibility roadmap for MG deployment, informing policymakers and investors to prioritize sustainable rural electrification in Ethiopia.

Keywords Ethiopia · GIS · Mini-hydro · Mini-grid · Solar PV · Wind

1 Introduction

Achieving universal electricity access, a key target of Sustainable Development Goal 7 (SDG 7), is a significant challenge in developing countries [1]. Traditionally, grid-based electrification has been the primary strategy for expanding electricity access [2]. However, in regions characterized by low population density, dispersed settlements, and limited infrastructure, grid extension is often impractical or infeasible [3]. In response to these challenges, decentralized energy systems, such as mini-grids (MGs) and standalone photovoltaic (SA PV) technologies, have emerged as viable alternatives [4].

MGs, which are powered by renewable resources such as solar, wind, and hydro, are increasingly favored to accelerate electricity access in off-grid areas [5]. They offer several advantages, such as rapid deployment and adaptability to local conditions [6]. Unlike centralized grid extensions, which require large upfront investments and long implementation

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periods, MGs are inherently scalable and can be installed incrementally as demand grows [7, 8]. In addition, they do not only provide basic lighting and electricity for households, but also support productive uses, such as the operation of agricultural machinery and small businesses, as well as social services like health centers and schools. This capability makes MGs stimulate broader socio-economic development in rural communities [9–11].

The recent increase in MG deployment is largely due to technological advancements and significant reductions in component costs [12]. For instance, between 2010 and 2022, the price of solar panels fell by 89%, while small wind turbine costs decreased by 49–64% in the same period. These trends have been further enhanced by advances in battery storage, which improve system reliability [13, 14]. These developments have made MGs increasingly affordable and scalable, making them a cost-competitive option for off-grid electrification. By 2022, about 21,500 MGs were supplying electricity to 48 million people, and projections suggest that by 2030, this number could increase to 29,400 MGs serving over 35 million people, predominantly in Africa and South Asia [15]. Nevertheless, while these statistics highlight the global momentum, they provide limited insight into which geographic locations or settlements would benefit the most, highlighting the need for precise, location-specific planning.

The success of MG deployment depends on identifying cost-optimal technologies and locations. This process requires a systematic, data-driven methodology to ensure both economic viability and long-term sustainability [16, 17]. Traditional site selection methods, which typically rely on stakeholder consultations and extensive field surveys, are often labor-intensive and hence not scalable to larger regions. In contrast, national planning strategies that use geospatial modeling and satellite data provide a more efficient framework [18]. Such approaches integrate a wide range of variables, streamline decision-making and ensure that investments are directed to projects with the highest likelihood of success.

The Geographic Information System (GIS) has become an essential tool in this endeavor, offering an unparalleled ability to combine and analyze a wide range of spatial datasets simultaneously and facilitating a multi-criteria decision-making process [19]. By combining data on resource availability—such as solar radiation, wind speed, and hydro potential—with factors such as population density, infrastructure proximity, terrain characteristics, and land use, GIS provides a comprehensive view of potential MG sites [20]. This spatial analysis allows policymakers and planners to prioritize sites that have not only technical potential, but also social and economic viability, ensuring that electrification efforts are aligned with broader development goals.

Although previous studies have made significant contributions to MG planning through GIS-based analysis, they have often focused on technical feasibility or cost comparisons with grid extension and solar home systems [2, 21–25]. For example, Korkovelos et al. [21], analyzed the technical feasibility of small/mini-hydro in sub-Saharan Africa (SSA) without assessing its economic feasibility. Similarly, Isihak [26] compared the cost of MG with grid extension and solar home systems, but neglected the social and environmental factors critical to project success. However, none of the aforementioned studies included these critical social and environmental factors that significantly influence the effectiveness of MG planning [27].

Incorporating social factors, such as population density, into GIS-based site selection offers notable advantages. First, it promotes equitable electricity access by targeting densely populated areas, thereby increasing transparency and fairness in resource distribution [19]. Second, population size directly influences electricity demand, a key determinant of MG cost-effectiveness [28]. Settlements with higher demand can justify larger investments, spreading fixed costs over more users and reducing per-unit costs. Moreover, environmental factors such as terrain slope, road access, and land cover data play an equally important role in minimizing installation and operating costs [29]. By integrating land-use data, GIS can exclude agricultural land, protected areas, urban built-up areas, forests, and water bodies, reducing land-use conflicts and displacement risks, even though MGs require a relatively small footprint compared to large-scale energy projects [30].

Community engagement complements this technical analysis, ensuring that MG projects are aligned with local priorities and realities. In countries like India, Nepal, Sri Lanka, Myanmar, and Indonesia, community engagement, such as collecting tariffs to fund operations, is essential to ensure financial sustainability [12]. Early discussions with residents can prevent land disputes and build support, thus aligning projects with local economic activities and cultural norms. This participatory approach not only increases acceptance but also leverages local knowledge to improve site selection, making MGs more responsive to the communities they serve.

Another critical consideration is the proximity of settlements to the existing grid infrastructure. In regions without clear policies or compatibility standards, grid expansion can overlap with MG sites, potentially rendering them obsolete [31]. This risk is particularly critical in countries with ambitious electrification plans, where grid lines may reach previously off-grid areas sooner than anticipated. To safeguard MG investments, site selection must prioritize locations that are unlikely to see grid arrival in the near future, guided by national electrification strategies. This strategic foresight will

ensure that MGs remain functional and relevant throughout their operational lifetime, providing sustainable benefits to remote populations.

Thus, the main objective of this study is to identify economically feasible locations for MG deployment using GIS-based multi-criteria decision-making frameworks. By improving our understanding of where economically feasible MGs can be deployed, this research aims to inform policy decisions, attract investment, and accelerate progress towards achieving universal electricity access. Specifically, this study addresses the following research questions:

- Which renewable energy-based MG offers the most cost-effective solution for rural electrification?
- What are the key spatial factors that influence the feasibility of MG deployment in underserved areas?
- How can GIS systematically integrate socioeconomic, infrastructure and resource data to identify economically viable locations for MGs?

1.1 Novelty of the research

This study provides a high-resolution, nationwide assessment of the economic potential for MG deployment in Ethiopia. While earlier research often emphasized technical feasibility or cost comparisons, this study integrates social and environmental dimensions, such as population density, terrain slope, road distance, and land cover, into the economic potential analysis. These often-overlooked factors are critical to reducing installation costs and ensuring project success. By mapping the spatial variability of renewable resources, this study provides a visual and data-driven guide to identify economically feasible MG locations.

2 Case study context: energy access and MG development in Ethiopia

The usefulness of the methodology scrutinized here is applied to Ethiopia to demonstrate its effectiveness in identifying feasible MGs locations. Ethiopia was chosen due to its need for sustainable electrification solutions. As Africa's second-most populous country, Ethiopia had a population of 125 million in 2022, yet only 55% had access to electricity, 43% in rural areas, leaving 56 million without electricity access [32]. This deficit ranks Ethiopia third globally in terms of absolute number, after Nigeria and the Democratic Republic of Congo [33]. Despite these challenges, Ethiopia is endowed with abundant renewable energy resources such as hydropower, solar, wind, and geothermal energy [34]. In contrast, biomass—predominantly wood and dung—remains the primary energy source for over 80% of households, highlighting the nation's heavy reliance on traditional fuels amid limited modern infrastructure [35].

Ethiopia's National Electrification Plan (NEP 2.0), launched by the Ministry of Water and Energy (MoWE), sets an ambitious goal of universal electricity access by 2025 through grid and off-grid strategies [36]. Under NEP 2.0, 65% of the population is targeted for connection to the national grid, leveraging the country's hydropower-dominated generation portfolio. The remaining 35%—consisting primarily rural and remote communities—will be served by off-grid solutions, such as solar SA PV systems and MGs [30]. In line with this plan, MoWE and the Ethiopian Electric Utility (EEU) have identified 250 potential MG sites in areas beyond the current reach of the grid, with a focus on PV-diesel hybrid systems under the Universal Electricity Access Program (UEAP). By 2021, 12 such MGs, with capacities ranging from 75 to 550 kWp, were already operational, thereby providing early evidence of the potential role of solar MGs in Ethiopia [37]. Although these initial deployments are modest, they indicate a promising trajectory for scaling up renewable MGs to meet rural energy needs.

International collaborations are also supporting these efforts. The Global Energy Alliance for People and Planet (GEAPP), in partnership with the African Development Bank and the European Investment Bank, is actively deploying solar-powered MGs across Ethiopia [38]. A flagship initiative in this context is the Distributed Renewable Energy–Agriculture Modalities (DREAM) project, which integrates MGs with agricultural infrastructure. Targeting smallholder farmers, the DREAM project powers irrigation systems, cold storage units, and food processing facilities to strengthen food security and enhance economic resilience. The project envisions that by 2030, 200 MGs will provide or expand access to electricity for more than 290,000 people and reduce greenhouse gas emissions by 200,000 tons. This represents a significant increase from the current solar MG capacity of 0.343 MW, which serves approximately 10,000 people as of 2022 [39]. This significant disparity between current capacity and future targets highlights the potential opportunities and challenges associated with scaling MG infrastructure to serve Ethiopia's vast rural areas.

Another factor driving the transition to renewable energy in Ethiopia is the rising cost of diesel. In July 2022, the Ethiopian government reduced fuel subsidies, resulting in a 30% increase in the price of gasoline and a 40% increase in the price of diesel. Over the past two years, diesel prices have increased by more than 200%, making diesel generators, once a staple for rural irrigation, increasingly unaffordable for farmers. This cost escalation is exacerbated by Ethiopia's reliance on imported fuel amid foreign exchange shortages. In response, the GEAPP's DREAM initiative is deploying solar-powered MG irrigation systems as a sustainable and cost-effective alternative to diesel-based solutions [38]. The limited reach of current MG deployments and the vast rural population still dependent on biomass underline the urgent need for strategic planning. This study's GIS-based approach aims to bridge this gap by identifying economically viable MG sites that align with NEP 2.0, ultimately advancing Ethiopia's pursuit of universal electrification.

3 Methodology

This study employs the Open-Source Spatial Electrification Tool (OnSSET) to assess the economically feasible off-grid mini-grids (MGs) in Ethiopia, focusing on mini-hydro, solar, and wind technologies. OnSSET is a GIS-based tool (framework) specifically developed for spatial electrification planning. It integrates high-resolution geospatial, socioeconomic, and renewable resource data to identify the least-cost electrification options at the settlement level [25]. By computing the Levelized Cost of Electricity (LCOE), OnSSET facilitates the determination of the most cost-effective option, be it grid extension, MG deployment, or SA PV systems—thus proving particularly effective for rural electrification assessments [40]. Its versatility is well documented in the scientific literature; the tool has been successfully applied at the regional level for SSA [2] and in country-specific rural electrification planning studies for Ethiopia [24], Nigeria [25, 41], Cameroon [42], Afghanistan [43], Burkina Faso [44], Kenya [45], Malawi [46], Bolivia [47], and Benin [48], underscoring its adaptability to different contexts.

The methodology consists of three key phases designed to systematically assess economically feasible MG locations. First, comprehensive geospatial and socioeconomic datasets are acquired, including renewable energy resource availability, grid infrastructure distributions, and population distribution. Second, feasibility criteria are established that incorporate technical, geographical, and socioeconomic factors. Third, economic feasibility is evaluated by calculating the LCOE for each renewable technology across settlements, thereby providing a geospatial benchmark for cost-effectiveness at the pre-feasibility stage. This structured, data-driven approach takes full advantage of OnSSET's capabilities to integrate technical, economic, and social dimensions into a cohesive framework for MG deployment.

3.1 Data acquisition

The analysis begins by collecting geospatial datasets, which are categorized into four main groups: renewable energy resources, geophysical parameters, infrastructure data, and environmental factors. As shown in Table 1, these datasets enable a thorough assessment of the economic potential of MG deployment. By integrating high-resolution spatial

Table 1 Summary of the key datasets used in the study

Data	Data source	Spatial resolution	Temporal coverage	Key parameters
Population settlement	Temesgen et al. [29]	–	2020	Population
Small/mini hydro	Korkovelos et al. [21]	–	2018	Hydropower potential
Solar	Global Solar Atlas [51]	250 m	1994–2018	GHI, DNI, and DHI
Wind	Global Wind Atlas [50]	250 m	2008–2017	Wind speed, power density
Land cover	Earth Data [58]	500 m	2001–2022	Land use classes
Slope	GDEM (NASA and Japan Space Systems) [60]	30 m	2009–2019	Elevation, slope
Power grid	EnergyData.info [70–72]	–	2012–2020	HV and MV lines, substations
Temperature	Global Solar Atlas [51]	1 km	1994–2018	Yearly average daily temperature
Road	Geofabric [73]	–	2018	Major roads, highways, and primary and secondary roads

information, the study captured Ethiopia's diverse geographic and socioeconomic landscape, thus establishing a basis for subsequent site selection and cost analysis.

3.1.1 Mini hydro

The potential for hydro MG in Ethiopia is assessed using a dataset developed by Korkovelos et al. [21], which covers the year 2018. This dataset leverages GIS-based models that incorporate runoff accumulation, elevation, and river network data to identify small- and mini-scale hydropower sites in SSA. In this study, we focus only on mini-hydro, defined as having capacities between 100 kW and 1 MW. These capacities are selected for their suitability in MG applications, providing sufficient power to meet the needs of rural communities [49].

Figure 1 illustrates the spatial distribution of mini-hydropower (100 kW–1 MW) and small-hydropower (1–10 MW) potential, showing a widespread presence across Ethiopia, with notable concentrations in the central and western regions. However, the reliability of the economic potential of hydro MG is subject to limitations inherent in the underlying datasets.

3.1.2 Solar

The economic potential of solar MGs is evaluated using high-resolution spatial solar radiation data provided by the Global Solar Atlas, which was developed by Solargis in collaboration with the World Bank's Energy Sector Management Assistance Program (ESMAP) [51]. This dataset provides long-term annual averages of daily totals for key solar parameters, including Global Horizontal Irradiance (GHI), Direct Normal Irradiance (DNI), temperature, and air quality, at a spatial resolution of approximately 250 m.

Figure 2a shows the distribution of GHI across Ethiopia, with values ranging from 1252 to 2350 kWh/m²/year. These data underscore Ethiopia's abundant solar resources, particularly in the northern and eastern regions. It should be noted, however, that because these values are derived from remote sensing, the use of annual averages for broad-scale analysis does not take into account for the intermittent nature of solar irradiance. Consequently, estimates of the economic potential of solar MGs that rely solely on these annual averages may either overestimate or underestimate system performance in specific locations, a limitation that is duly acknowledged.

3.1.3 Wind

The feasibility of wind MG is analyzed using annual average wind speed data from the International Renewable Energy Agency (IRENA) Global Atlas for Renewable Energy, which is produced by the Technical University of Denmark (DTU) in collaboration with the World Bank Group [50]. The dataset provides wind speed measurements at several heights above ground level (10, 50, 100, 150, and 200 m) with a spatial resolution of 250 m.

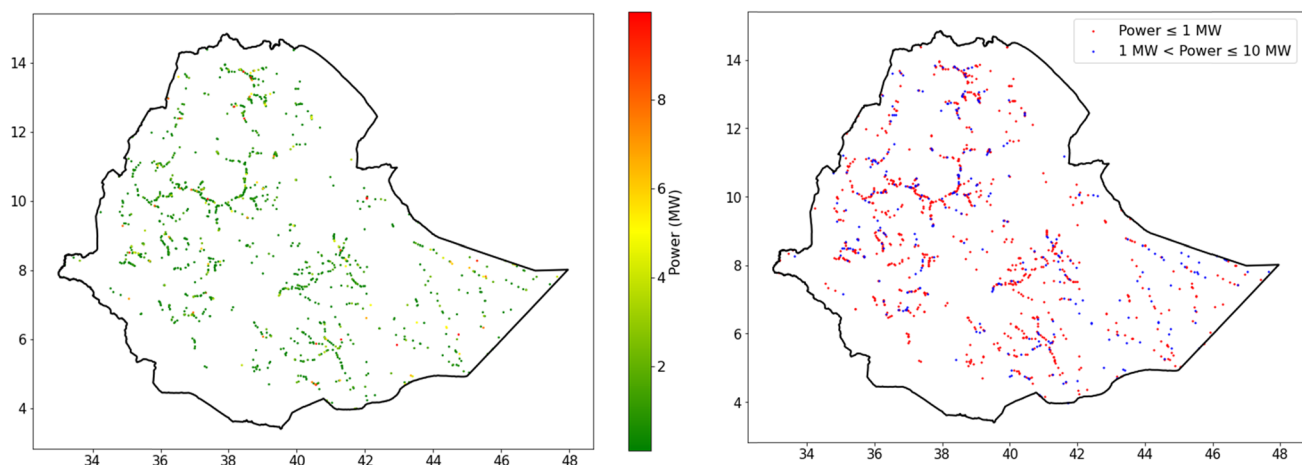


Fig. 1 Spatial distribution of mini- and small-scale hydropower potential across Ethiopia [21]

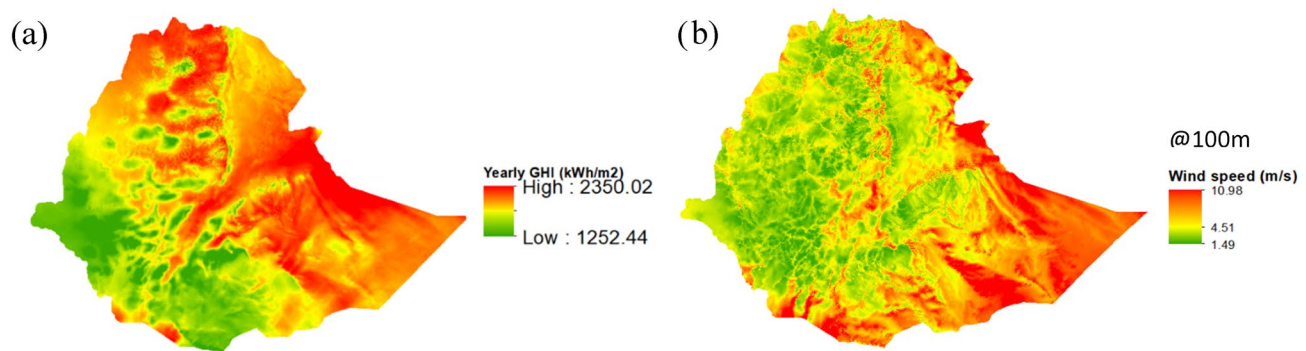


Fig. 2 Spatial distribution of solar and wind energy resources in Ethiopia: **a** GHI (left), and **b** wind speed at 100 m (right) [51, 52]

Figure 2b depicts the wind speed distribution at 100 m above surface level. It highlights that the eastern regions of Ethiopia experience higher wind speeds, which are favorable for wind power generation. Similar to the solar dataset, relying on annual averages for wind speeds introduces potential inaccuracies because wind resources can exhibit significant temporal variability due to seasonal changes and local weather dynamics. This intermittency, which is not fully captured by the average values, may influence turbine performance and thereby affect the economic viability of wind MGs, a factor that is not explicitly incorporated into the LCOE assessment.

3.2 Criteria for economic feasibility assessment

To evaluate the economic feasibility of off-grid MGs in Ethiopia, the study integrates several key decision criteria. These include land cover suitability, terrain slope, proximity to infrastructure, population density, and technology-specific constraints, to identify cost-effective and sustainable MG locations.

The rationale for selecting these criteria is based on the literature and tailored to the specific context of Ethiopia [29, 53]. These criteria were chosen due to their direct influence on MG's installation costs, operational feasibility, and long-term viability. All factors were treated as equally important (having equal weights), with the assumption that all are integral to the success of MG projects. For example, even if a site has abundant wind resources, steep and inaccessible terrain could result in prohibitive construction and transportation costs; similarly, high population density alone does not compensate for unsuitable land conditions. These filters ensure that only settlements that meet all criteria simultaneously are deemed viable for MG deployment.

3.2.1 Population density

Population density is a critical determinant of the economic viability of MGs, as it directly affects electricity demand, which in turn justifies capital investment. Deploying MGs in locations with very few households can result in higher unit costs, while larger populations allow fixed costs to be spread over a larger number of users, thereby increasing cost-effectiveness. NEP 2.0 suggests a minimum of 250 households for MG viability [34], but this study uses a threshold of 300 households, validated by the successful solar PV MG in Omorate, southern Ethiopia [54].

The High-Resolution Settlement Layer (HRSL), with an approximate resolution of 1 arc-second (about 30 m at the equator), provides detailed population distribution estimates. This data is used to delineate settlements using OnSET's clustering algorithm [55, 56]. The processing step involves clipping the HRSL data to administrative boundaries, eliminating low-density cells, merging eight contiguous raster cells into clusters, and calibrating to match the base year population (2022) (see [55]). The clustering of Ethiopia resulted in 809,087 population settlements/clusters [29]. Despite its accuracy in estimating population distribution, HRSL has limitations due to its reliance on satellite imagery and historical census data, which may not adequately capture the rapid population changes occurring in urbanizing areas [57].

3.2.2 Land cover suitability

Ensuring suitable land cover is essential to minimize environmental barriers when deploying MG infrastructures. In this study, land cover types such as open shrublands, savannas, grasslands, and barren lands, are considered suitable

for MG deployment, while areas dominated by dense forests, wetlands, or designated protected zones are excluded from the analysis. Land cover data was obtained from Earth Data [58], which provides global land cover data at a resolution of 500 m and has been available since 2001 [59]. Figure 3 shows Ethiopia's land cover (as of 2020), classified into 15 different categories based on MODIS/Terra + Aqua data.

3.2.3 Terrain slope

Terrain slope is another important factor in determining suitable locations for MG deployment. Steep slopes can complicate the construction, installation, and ongoing maintenance of energy infrastructures. For example, slopes of less than 10 degrees allow for the safe installation and operation of equipment such as solar panels, wind turbines, and hydroelectric generators [46]. The Advanced Spaceborne Thermal Emission and Reflection Radiometer Global Digital Elevation Model (ASTER GDEM), with a resolution of 30 m, is used to map Ethiopia's terrain [60]. As shown in Fig. 4, most Ethiopian settlements lie below the 10-degree terrain slope threshold.

3.2.4 Road distance

Road accessibility plays a key role in determining the economic viability of MG projects, as it directly impacts transportation costs for materials, equipment, and spare parts. Therefore, in this study, settlements located within 1 km of main roads are considered for MG deployment [53]. This threshold was chosen to ensure efficient material delivery, facilitate routine maintenance, and enhance overall operational efficiency and cost effectiveness.

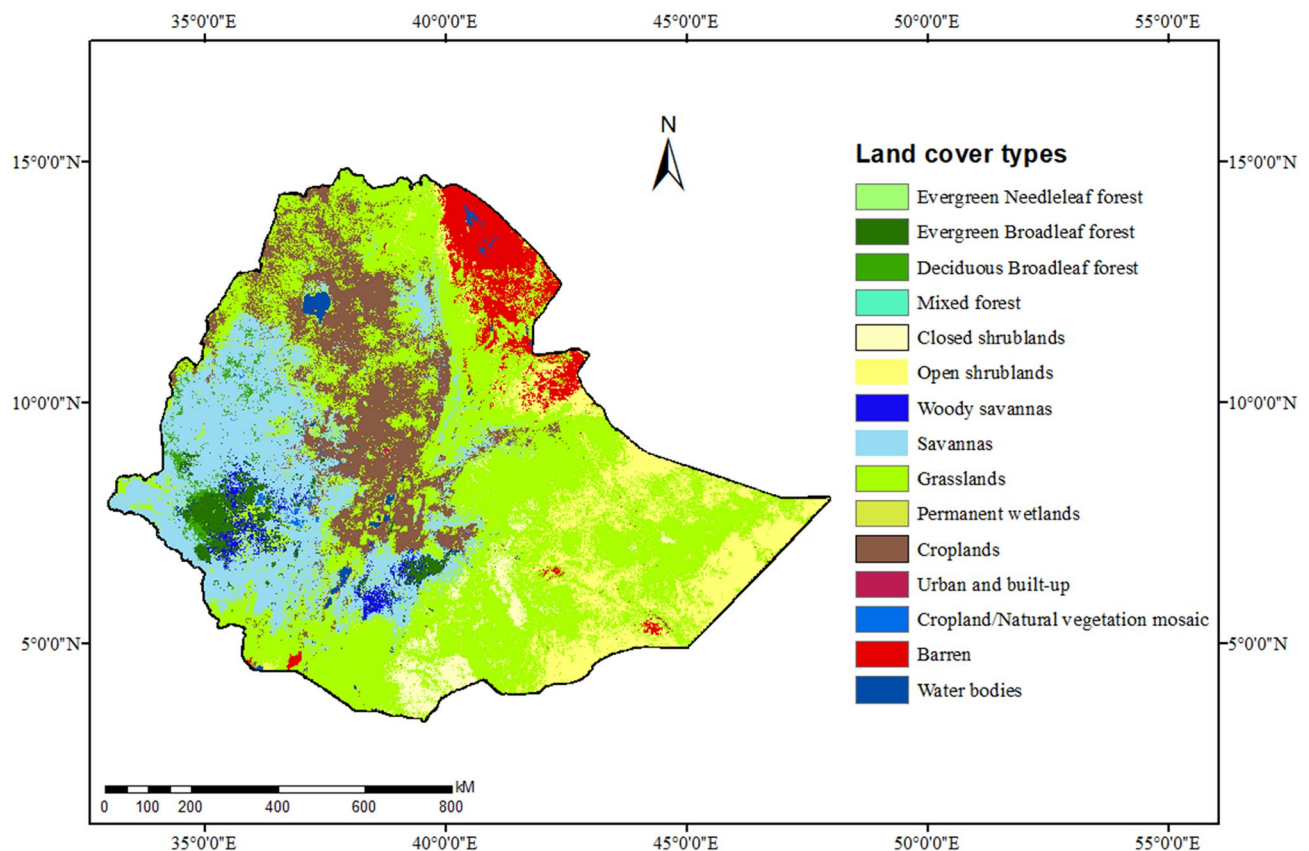


Fig. 3 Land cover types in Ethiopia based on MODIS Terra/Aqua data for 2022 [29]

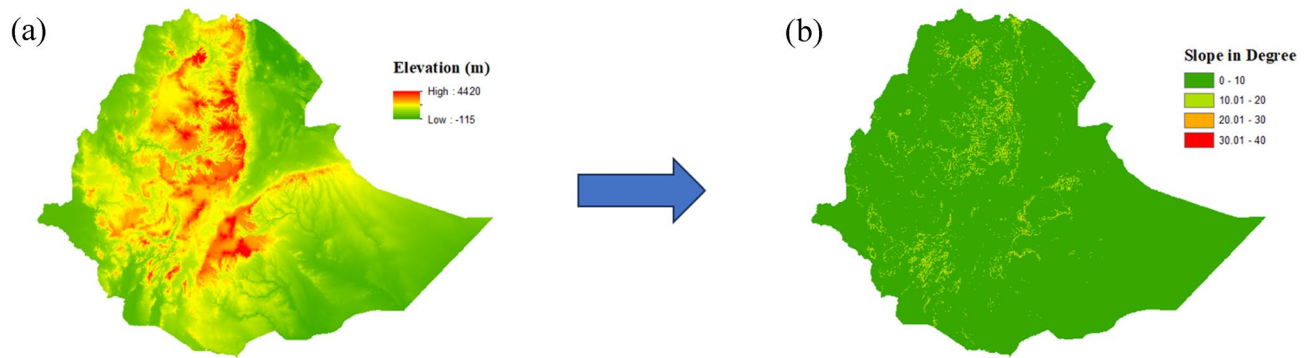


Fig. 4 Topographic maps of Ethiopia (2020): **a** elevation map (left) and **b** derived terrain slope map (right) [60]

3.2.5 Power grid infrastructure

Assessing the proximity of potential MG sites to existing power infrastructure is a critical step in ensuring the long-term viability of off-grid electrification investments. Sites located too close to the current grid network face a high risk of future grid extension, which could render MG systems obsolete, resulting in stranded assets and operational inefficiencies. Therefore, spatial exclusion based on grid proximity is essential to avoid redundancy and to ensure that MG deployment targets settlements genuinely beyond the foreseeable reach of grid expansion.

NEP 2.0 provides strategic guidance for such exclusions. It prioritizes grid intensification within 2.5 km of existing medium-voltage (MV) lines, aiming for near-term connections in these areas. Additionally, NEP 2.0 envisages that up to 96% of the population may eventually be connected to the grid if located within 25 km of MV lines [34]. This framework underscores the cost-effectiveness of leveraging existing infrastructure for grid extension, particularly in densely populated or logistically accessible areas. To align with these policy directives and minimize the risk of infrastructure overlap, this study defines two exclusion scenarios based on grid proximity.

- 2.5 km grid intensification scenario:** This scenario excludes settlements within 2.5 km of existing MV lines. The rationale is that these areas are prioritized for grid connection under NEP 2.0 due to their proximity to MV lines. Deploying MGs here would be shortsighted, as the grid infrastructure is likely to eventually replace or displace them. Thus, excluding these settlements ensures that MG deployment focuses on areas where off-grid solutions can provide sustainable benefits.
- 25 km grid intensification scenario:** This scenario excludes settlements located within 25 km of existing MV lines. Although these areas may not be imminently electrified, they remain within the strategic horizon for future grid integration. This larger buffer allows for economic and logistical barriers, such as high transmission costs across Ethiopia, that may delay grid reach, making MGs a practical interim or permanent solution. Consequently, settlements beyond 25 km, which are often remote or sparsely populated, are prioritized for MG deployment as they are less likely to benefit from grid access in the foreseeable future.

Figure 5 presents the spatial distribution of existing high-voltage (HV) infrastructure and substations, along with projected MV lines, derived from our recent study [29]. This mapping supports MG site selection by identifying settlements beyond both grid proximity thresholds, ensuring alignment with national electrification priorities and enhancing the durability and impact of off-grid systems.

3.2.6 Technology-specific constraints

Each renewable energy technology has unique constraints that refine the economic viability assessment.

3.2.6.1 Mini-hydro criteria The selection of sites for hydro MG is limited by their proximity to settlements and their hydrological performance. Only hydro sites within 50 km of population settlements are considered viable, a threshold that is consistent with the commonly assumed distance for MV line extensions in rural electrification planning, as high-

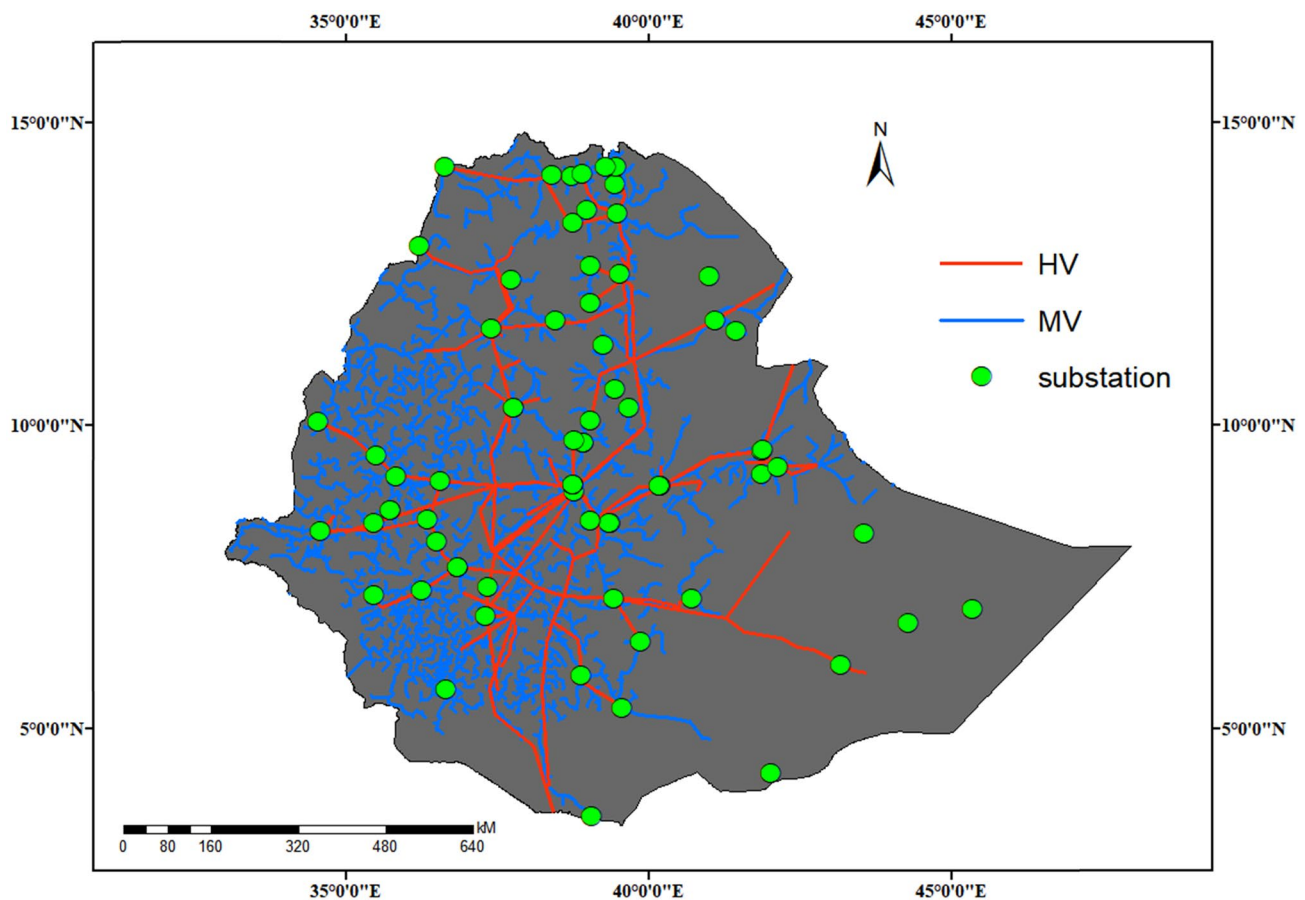


Fig. 5 Existing HV and substations, and projected MV lines across Ethiopia [29]

lighted by Mentis et al. [24]. Mini-hydro sites farther than 50 km from population settlements are deemed unfeasible due to increased costs associated with transmission line extensions and infrastructure maintenance.

Another critical parameter influencing the viability of MG hydropower is the capacity factor (CF¹), which depends on site-specific hydrological conditions, design parameters (e.g., exceedance levels), and operational characteristics (e.g., dispatchable systems or run-of-river configurations) [61]. In the literature, CF values for small- and mini-hydropower vary from 25–46% [31, 62]. In Ethiopia, where seasonal rainfall drives river variability, lower CFs may occur in drier regions, introducing uncertainty into the economic analysis. Therefore, to capture this variability, the economic potential of MG hydro is assessed using CF scenarios of 25 and 46%.

3.2.6.2 Solar PV criteria The feasibility of solar PV MG is highly dependent on the GHI values, while the conversion efficiency of the PV panel is affected by the ambient temperature [63]. Higher temperatures, which are common in the Ethiopians lowlands, reduce panel efficiency due to heat-related losses. To account for this, the study adjusts the efficiency using a temperature coefficient of $-0.5\%/^{\circ}\text{C}$, which is typical for silicon cells that experience efficiency loss above the Standard Test Condition (STC) temperature of 25°C [64]. The adjusted efficiency is calculated as expressed in Eq. 1.

$$\eta_{adjusted} = \begin{cases} \eta_{STC} \times (1 + \beta \times (T - 25)), & T \geq 25^{\circ}\text{C} \\ \eta_{STC}, & \text{otherwise} \end{cases} \quad (1)$$

¹ CF is the actual energy output of an electricity-generating device divided by the energy output that would be produced if it operated at its rated power output.

where $\eta_{adjusted}$ is the adjusted efficiency, η_{STC} is the efficiency at STC (25 °C), β is the temperature coefficient ($-0.5\%/^{\circ}\text{C}$), and T is the ambient temperature in degrees Celsius. This adjustment accounts for temperature-related losses in panel efficiency and ensures realistic performance estimates across Ethiopia's climatic zones, from the cool highlands to the arid eastern regions.

3.2.6.3 Wind power criteria Wind MG potential depends on local wind speeds, with turbines generating electricity between 3 and 25 m/s [65]. A cut-in speed of 4 m/s is established as the minimum threshold for viability, ensuring that energy production justifies the associated costs [66]. The study uses the Vestas V44 wind turbine, with a nominal capacity of 600 kW and a hub height of 55 m. The selection of wind turbine with a hub height of 55 m is a practical decision based on the typical specifications of small-scale wind turbines used in MGs [67–69]. In Ethiopia, large-scale wind turbines typically use hub heights below 85 m, and small-scale off-grid wind turbines often operate at even lower hub heights. These turbines are more suitable for local, off-grid applications.

Given that the wind speed data were originally measured at a height of 100 m, it was necessary to adjust these values to reflect wind speeds at 55 m, the hub height of the selected wind turbine. This adjustment was performed using the logarithmic wind profile law, which accounts for the variation in wind speed with height due to changes in surface friction. The adjusted wind speed at 55 m (U_z) was computed using Eq. 2, based on the vertical wind shear relationship. As wind speed generally increases with height, this correction ensures a more accurate estimation of turbine performance under real-world conditions. Figure 6 illustrates the power curve of the selected 600 kW wind turbine, demonstrating its energy generation capabilities at varying wind speeds.

$$U_z = U_r \left(\frac{z}{z_r} \right)^{\alpha} \quad (2)$$

where U_r is the wind speed at measured height (100 m), z is the desired hub height (55 m), z_r is the height at which the wind speed is measured, and α is the wind shear exponent, which accounts for how wind speed changes with height and is calculated based on the surface roughness. This exponent is location-specific and the OnSSET model calculates it based on the wind speed and hub height, ranging from 0.19 to 0.31, in Ethiopia depending on the location. The OnSSET model calculates the wind CF, as shown in Eq. 3.

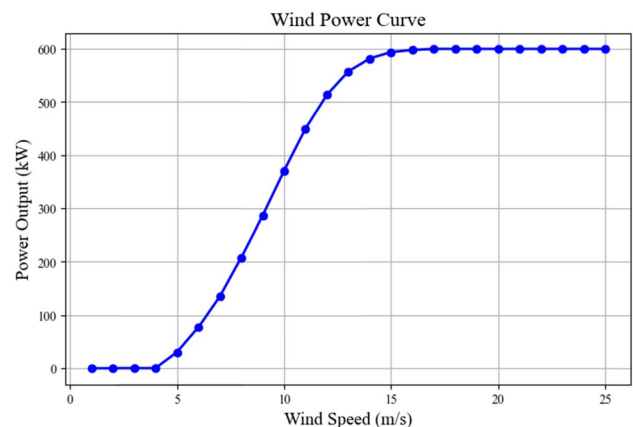
$$CF = \frac{E_{expected}}{P_{rated} * T} \quad (3)$$

where $E_{expected}$ represents the expected energy output (adjusted for a 0.97 availability factor and 0.85 loss factor), P_{rated} is the rated capacity of the turbine (600 W), and T is the total period (8760 h).

3.3 LCOE

The economic viability of off-grid MGs is assessed using the LCOE, a widely used metric that calculates the present-value cost of producing 1 kilowatt-hour of electricity over the life of a project [63, 74]. In this study, LCOE is used as a decision-support method to assess the cost-effectiveness of MG deployment across candidate settlements. Once suitable

Fig. 6 Power curve of the Vestas V44 wind turbine with a rated capacity of 600 kW



settlements are identified based on economic viability criteria, the LCOE is calculated by dividing the discounted lifetime costs by the discounted lifetime energy generation. This metric incorporates all financial aspects, including capital expenditure (CAPEX), operations and maintenance (O&M) costs, and cumulative energy generation over the project's lifetime [40]. The LCOE formula used in this analysis is as expressed in Eq. 4.

$$LCOE = \frac{\sum_{t=1}^n \frac{I_t + O\&M_t}{(1+r)^t}}{\sum_{t=1}^n \frac{E_t}{(1+r)^t}} \quad (4)$$

In this equation, I_t represents capital expenditure in year t , $O\&M_t$ denotes the operation and maintenance expenditure for a given technology in year t , E_t signifies the electricity generation in year t . The annual energy output (kWh/year) was calculated for each location based on the annual available resources provided in Figs. 1, 2a, b. Here, r is the discount rate used to calculate the present value of future costs and benefits, and n is the expected lifetime of a project, which varies by technology.

In this study, an 8% discount rate is assumed, reflecting the time value of money and potential risks associated with future cash flows. This rate aligns with standard energy project evaluations in developing contexts [46, 67, 75].

3.3.1 Technoeconomic data and key assumptions

Accurate LCOE calculations rely on robust techno-economic data tailored to Ethiopia's off-grid MG context. Capital costs, O&M costs, CFs, and project lifespans for hydro, solar, and wind MGs are taken from peer-reviewed studies focused on off-grid MG technologies in Ethiopia and other developing countries [14, 42, 44, 76]. This ensures that the data reflect real-world conditions and are applicable to the local context, improving the relevance and reliability of the results.

For hydro MGs, capital costs are estimated at 3,000 \$/kW, which includes turbines, and civil works (e.g., dams and penstocks). Solar PV MGs are estimated to have a capital cost of 2,950 \$/kW, which includes photovoltaic panels, inverters, mounting structures, and balance-of-system components. Wind MGs incur the highest capital cost, estimated at 3,750 \$/kW due to the complexity of installing turbines in remote areas [14, 76].

O&M are modeled as an annual percentage of the initial capital investment. Hydro MGs incur higher O&M costs (3% of capital cost per year) due to the mechanical complexity of the turbines and the need for regular sediment removal. Solar and wind MGs, require lower annual O&M expenditures (2% of capital cost), primarily for panel cleaning, inverter maintenance, and turbine inspections [44].

Capacity factors are also technology-specific. For hydro-based MGs, two representative CF scenarios are considered: a lower bound of 25% and an upper bound of 46%, as described in Sect. 3.2.6.1. For solar and wind MGs, CFs are calculated based on the annual resource availability in each settlement (based on Fig. 2a, b), using the OnSSET model [31]. This geospatial model calculates the CF of solar MG by analyzing the solar GHI and using Eq. 2 for wind MGs. The CF is 19–28% for solar PV, and 10–47% for wind MG across the entire country.

Project lifespans are assumed to be 30 years for hydro MGs, owing to the durability of hydropower infrastructure, and 20 years for solar and wind MGs, in line with global averages for renewable energy technologies [42] (Table 2).

3.4 Sensitivity analysis

Given the uncertainties in the discount rate assumptions used to calculate the LCOE calculations, a sensitivity analysis is conducted. The discount rate is a critical parameter influencing the present-value cost of MG investments and,

Table 2 Electricity generation technology parameters used in the model [14, 42, 44, 76]

MG technology	Capital cost	O&M cost (% investment/year)	Capacity factor (%) [*]	Lifespan (years)
Mini-hydro	3000 \$/kW	3	25–46	30
Solar	2950 \$/kW	2	19–28	20
Wind	3750 \$/kW	2	10–47	20

^{*}Capacity factors for solar and wind technologies are calculated based on the annual resource availability in each settlement by the OnSSET model

consequently, their economic feasibility. Three different discount rates are analyzed: a low-cost capital with a discount rate of 5%, a medium-cost capital with a discount rate of 8% (which serves as the base case), and a high-cost capital with a discount rate of 12%. Such comparisons of different discount rates are intended to show the economic feasibility of MGs under different cost-of-capital conditions [62].

4 Results and analysis

This study evaluates the economic feasibility of mini-hydro, solar PV, and wind MGs across Ethiopia, using LCOE estimates under two grid proximity scenarios (2.5 and 25 km from MV lines). These findings, based on the methodology outlined in Sect. 3, highlight the potential of each technology to address Ethiopia's rural electrification gap, guided by LCOE estimates.

4.1 Hydro MG economic potential

The findings reveal that suitable mini-hydro sites for MG deployment are widely distributed under the 2.5 km grid intensification scenario. In contrast, under the 25 km grid intensification scenario, the identified sites are predominantly concentrated in the eastern part of the country. This pattern corresponds to settlement distribution in eastern Ethiopia, where many communities are located more than 25 km from MV lines. In both scenarios, the hydro sites exhibit power generation capacities ranging from 0.1 to 0.99 MW.

Specifically, the study identifies 306 hydro MG sites under the 2.5 km scenario and 84 sites under the 25 km scenario. These sites could potentially electrify approximately 5 million people under the 2.5 km scenario and 1.9 million people under the 25 km scenario. Figure 7 provides a visual representation of the spatial distribution of these resources, taking into account factors such as land cover suitability, terrain slope, population density, and proximity to existing MV grid lines and roads as explained in Sect. 3.2. Furthermore, the LCOE analysis shows that hydro MG could generate electricity at costs ranging from 0.09 to 0.16 \$/kWh for CF of 46% (optimal flow) to 25% (seasonal low flow), respectively.

4.2 Solar MG economic potential

The analysis of solar MG economic potential reveals distribution patterns similar to those observed for hydro MGs. Under the 2.5 km grid intensification scenario, solar MG sites are widely dispersed, capitalizing on Ethiopia's consistently sunny climate. In contrast, the 25 km scenario shows a pronounced concentration of sites in eastern Ethiopia, a region where settlements are generally remote and less likely to be connected to the grid (see Sect. 3.2.5).

The annual average solar radiation in Ethiopia ranges from 1,643 to 2,446 kWh/m²/year, with the highest values observed in the northern and eastern lowland regions. Under the 2.5 km scenario, the analysis identifies 1,665 suitable settlements that could potentially serve approximately 7.2 million people. Conversely, the 25 km scenario yields 398 suitable settlements, covering a population of about 3.2 million people. This contrast illustrates how the broader reach

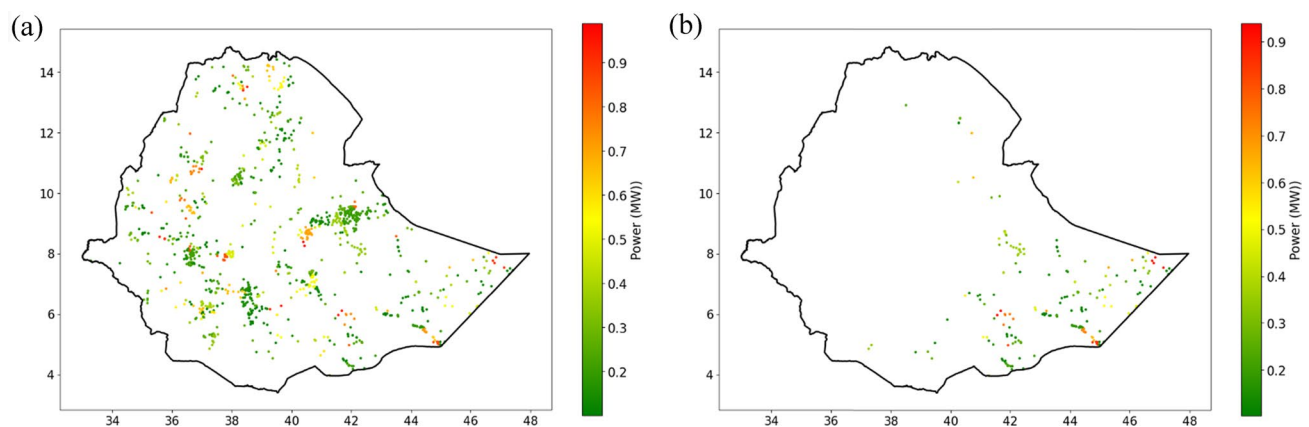


Fig. 7 Spatial distribution of potential mini-hydro MG (in MW): **a** under the 2.5 km grid intensification scenario (left) and **b** under the 25 km grid intensification scenario (right)

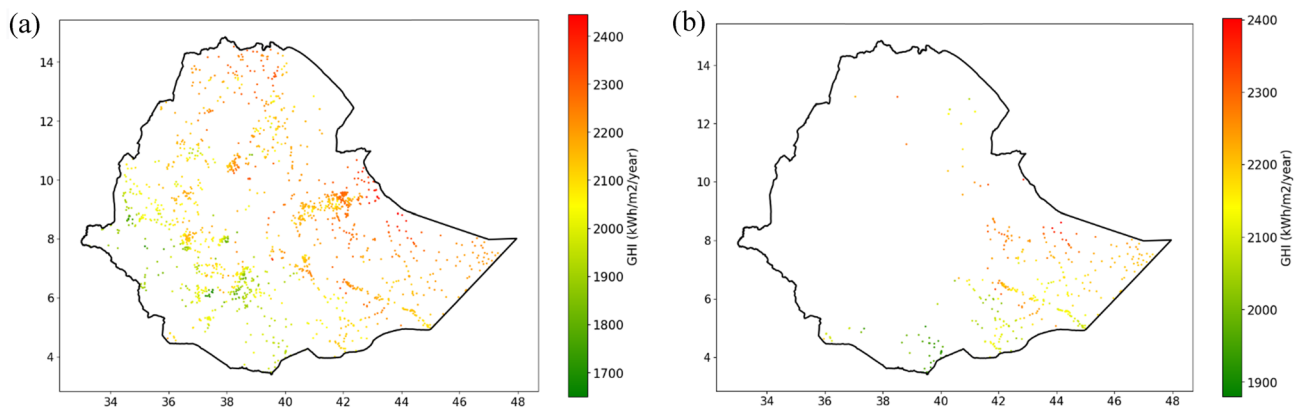


Fig. 8 Spatial distribution of solar PV MG energy potential (in kWh/m²/year): **a** under the 2.5 km grid intensification scenario (left) and **b** under the 25 km grid intensification scenario (right)

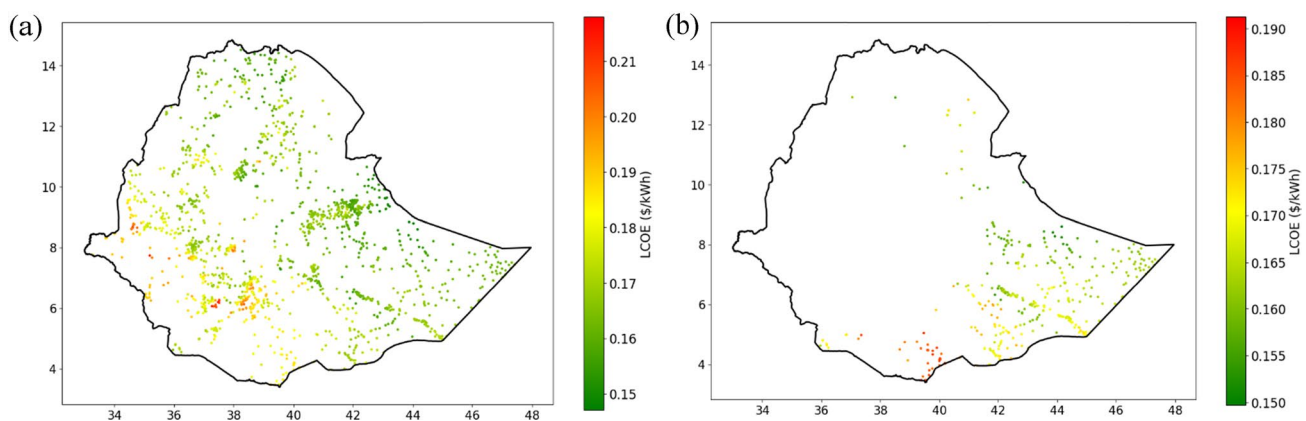


Fig. 9 Spatial variation in LCOE for solar MGs: **a** under the 2.5 km grid intensification scenario (left) and **b** under the 25 km grid intensification scenario (right)

of the 2.5 km scenario takes advantage of the favorable solar resource, while the 25 km scenario is more relevant for remote areas where off-grid solutions are critical.

Furthermore, the mapping in Fig. 8 incorporates temperature-adjusted efficiency (see Sect. 3.2.6.2), which accounts for reductions in performance in regions with elevated ambient temperatures. This adjustment ensures that the estimated energy generation more accurately reflects the variation in performance across Ethiopia's diverse climatic zones.

The LCOE analysis of the solar MGs indicates values ranging from 0.15 to 0.22 \$/kWh. In particular, over 75% of the evaluated settlements have LCOE values below 0.18, and 0.16 \$/kWh under the 25 and 2.5 km grid intensification scenarios, respectively. On average, the estimated LCOE is approximately 0.17 \$/kWh. Figure 9 shows the economic potential of solar MGs by mapping these LCOE estimates, as calculated using the methodology described in Sect. 3.3.

4.3 Wind MG economic potential

The economic potential for wind MGs exhibits a spatial trend similar to that of solar MGs. In the 2.5 km scenario, the wind MG potential is widespread, while in the 25 km scenario, it is concentrated in the eastern regions, where wind speeds are highest. All identified settlements can generate wind energy ranging from 155 to 2200 MWh/year.

Specifically, the assessment identifies 922 settlements with viable wind MG potential under the 2.5 km scenario, compared to 369 settlements under the 25 km scenario. These suitable areas could potentially serve approximately 4.8 million people in the 2.5 km scenario and about 3.1 million people in the 25 km scenario. Figure 10 illustrates the spatial distribution of the MG wind potential in both scenarios.

The LCOE analysis for wind MGs in Ethiopia indicates a broad range of values, from 0.12 to 1.75 \$/kWh. Notably, over 75% of the settlements evaluated have LCOE values below 0.94 and 0.56 \$/kWh under the 25 and 2.5 km grid

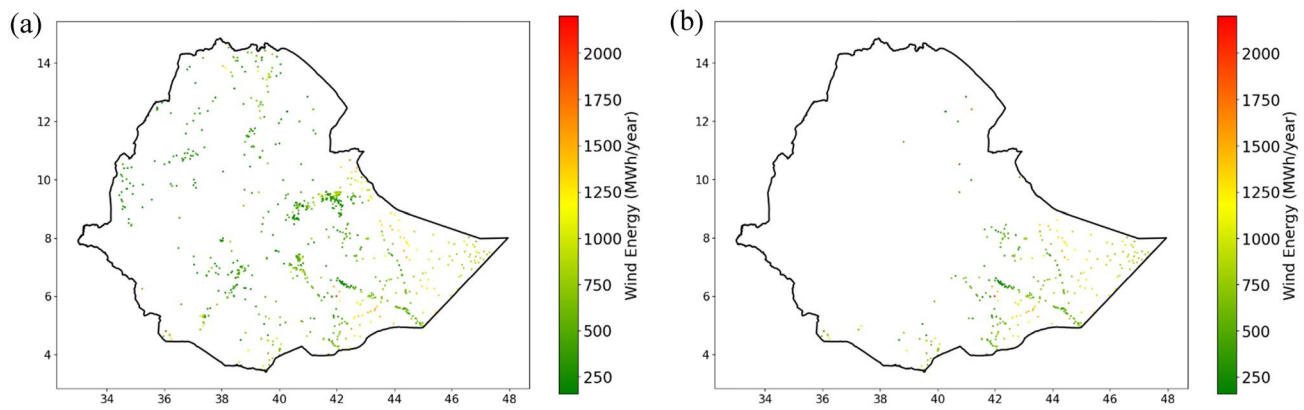


Fig. 10 Spatial distribution of MG wind energy potential (in MWh/year): **a** under the 2.5 km grid intensification scenario (left) and **b** under the 25 km grid intensification scenario (right)

intensification scenarios respectively. The average LCOE is estimated to be 0.69 \$/kWh. Figure 11 presents this economic potential through the spatial distribution of LCOE values for potential wind MGs across Ethiopia.

Comparative economic analysis indicates that even in regions with optimal wind conditions, wind MG is generally less cost-competitive than solar and hydro MG alternatives. This is primarily due to the higher capital costs associated with wind energy and its lower capacity factor, which is affected by intermittency and terrain-specific wind conditions. Yet, it remains viable in the windy eastern region and complements a diversified MG portfolio. Table 3 provides a comparative summary of the three renewable MG technologies, highlighting their respective economic viability, population reach, and regional suitability.

4.4 Sensitivity analysis

The sensitivity analysis shows how different discount rates (5, 8 and 12%) affect the LCOE of each MG technology. When the discount rate is reduced from the base case of 8% to 5%, the LCOE for hydro MGs decreases by approximately 16–20%.

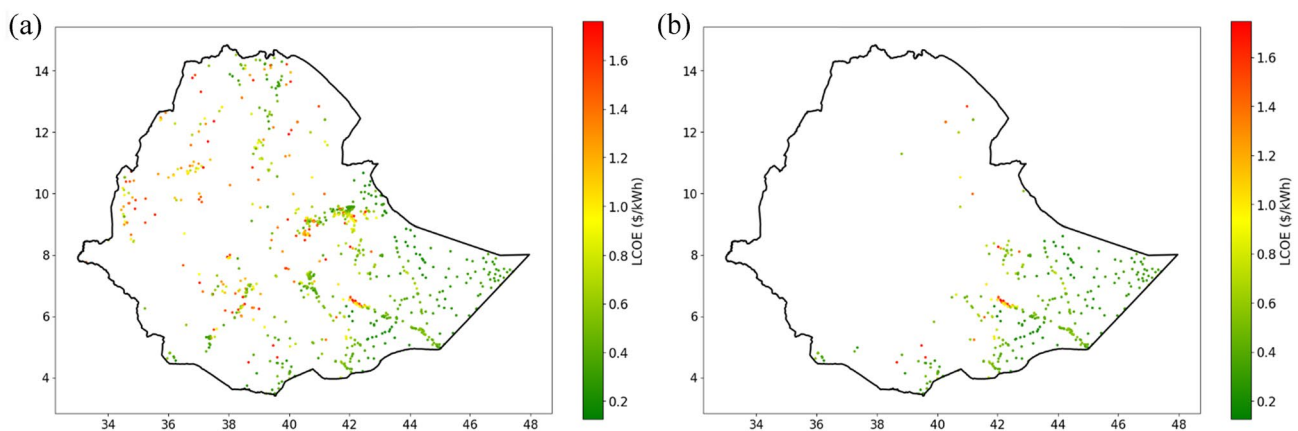


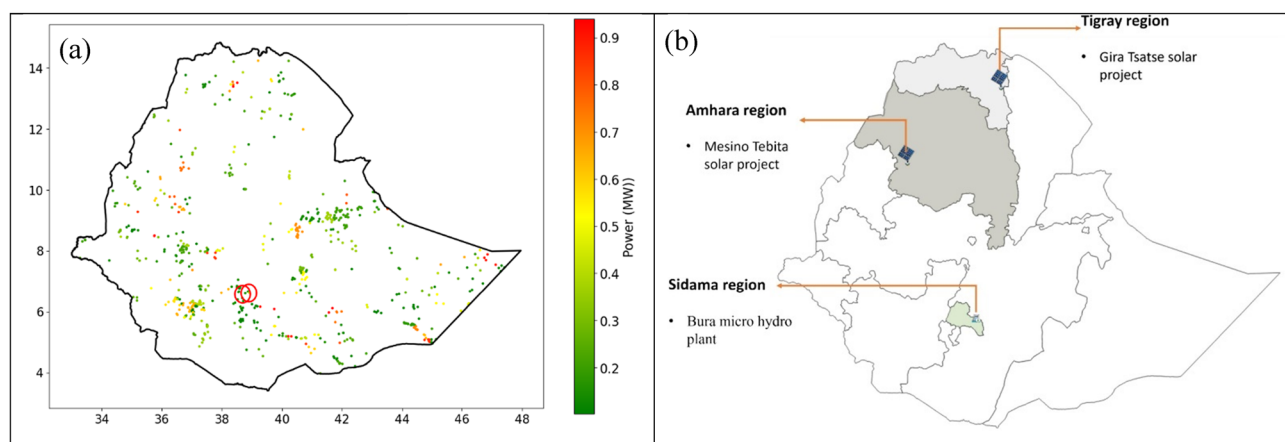
Fig. 11 Spatial variation in LCOE for wind MGs: **a** under the 2.5 km grid intensification scenario (left) and **b** under the 25 km grid intensification scenario (right)

Table 3 Summary of key findings

MG technology	LCOE range (\$/kWh)	Population to be served (Million)	Regions
Hydro	0.09–0.16	5	Western/Central
Solar	0.15–0.22	7.2	Northern/Eastern
Wind	0.12–1.75	4.8	Eastern/Northern

Table 4 Sensitivity analysis

MG technology	Discount rate of 5%	Discount rate of 8% (base case)	Discount rate of 12%
Hydro	0.071–0.13	0.09–0.16	0.11–0.21
Solar	0.12–0.18	0.15–0.22	0.18–0.27
Wind	0.10–1.45	0.12–1.75	0.15–2.2

**Fig. 12** Validation of hydro MG sites: **a** Economically feasible sites identified in this study (left) and **b** Existing micro-hydro systems in Ethiopia (right [30])

Conversely, increasing the discount rate to 12% increases the LCOE for hydro MG by 25–31% relative to the base case. A similar pattern is observed for solar MGs: a 5% discount rate results in an 18–20% decrease in LCOE, while a 12% rate results in a 20–23% increase. For wind MGs, the LCOE decreases by 16–17% at a 5% discount rate, but increases by 25–26% when the discount rate is increased to 12% compared to the base case.

These results underscore that all three technologies are significantly influenced by financial parameters. Lower discount rates reduce the present value of future costs, thereby improving economic viability, whereas higher discount rates increase the LCOE and may undermine project feasibility. The detailed impact of discount rate variations on the economic viability of MGs is summarized in Table 4.

4.5 Validation

To ensure the reliability and applicability of this GIS-based assessment, the study validates its results by comparing them with existing renewable energy projects in Ethiopia, thereby strengthening the credibility of the methodology. Given the limited number of operational MGs in Ethiopia, particularly for hydro and wind, our validation approach involved comparing our results with the following: a) existing community energy micro-hydro systems (for MG hydro validation), b) operational solar PV MGs in Ethiopia (for solar MG validation), and c) utility-scale wind farms (for wind MG validation).

For the hydro MG results, we compared our findings with existing community energy micro-hydro plants. While there are no known hydro MG projects, we identified Bura and Gowacho II micro-hydro plants in the Sidama region as relevant benchmarks [30]. These sites, developed under community energy initiatives, closely match with the hydro MG locations identified in this study, particularly in the 2.5 km scenario.

Figure 12 juxtaposes our economically feasible sites with these micro-hydro plants, revealing a strong spatial overlap. This correspondence validates the site selection criteria (Sect. 3.2), such as proximity to rivers and settlements, reinforcing the methodology's ability to pinpoint viable hydro MG locations despite limited operational precedent.

The solar MG results are validated against Ethiopia's 12 operational solar PV MGs, documented in 2021 [37]. Our analysis identifies sites such as Omorate, Werder (Dollo), Behima, Beltu, Ungoge, Albasa, Benbaho, and Arae as economically feasible, mirroring these active installations. Figure 13 illustrates this alignment, with most of the identified sites matching the operational MG locations, confirming the accuracy of the criteria used such as GHI and population density (Sect. 3.2). However, there are discrepancies with Tum and Kusrewad MGs, which are excluded due to their proximity within 2.5 km

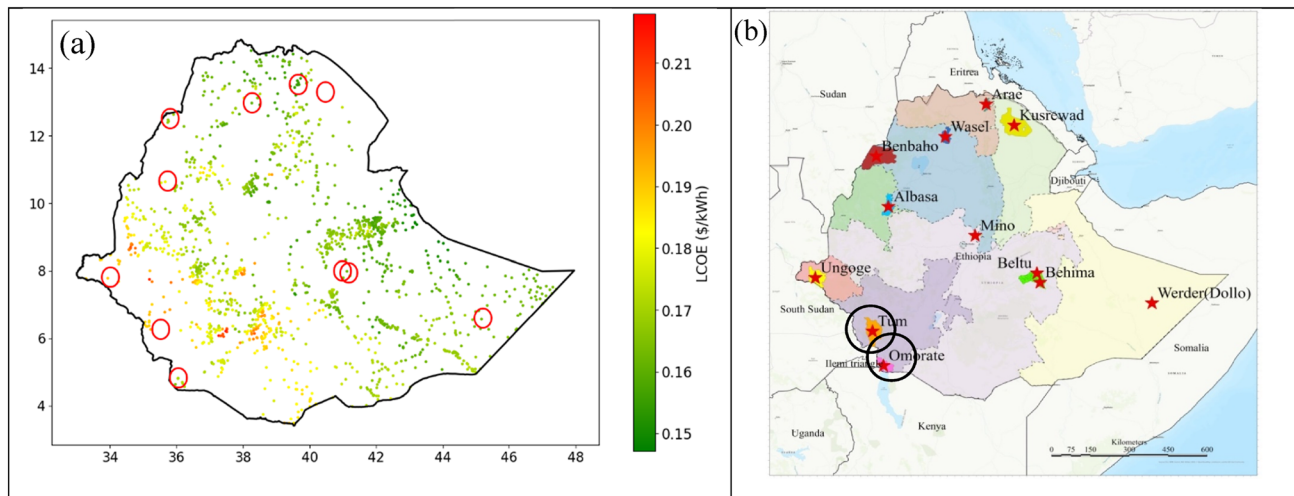


Fig. 13 Validation of solar MG sites: **a** Economically feasible solar MG sites identified in this study (left) and **b** Existing 12 solar PV MGs in Ethiopia (right [37])

of MV lines, a criterion that prioritizes grid connection. Additionally, the Mino MG could not be validated due to missing GPS data in public records. Despite these outliers, the broad agreement between our findings and deployed systems underscores the practical relevance of the methodology for solar PV MG planning.

The potential for wind MGs was validated against Ethiopia's utility-scale wind farms: Ashegoda, Asysa, and Adama [77]. While these are larger than typical MGs, their locations in high-wind eastern and northern regions coincide with our identified sites, especially in the 2.5 km scenario (Fig. 14). This overlap validates the wind speed threshold (≥ 4 m/s), suggesting that wind MGs can thrive in similar zones. The eastern concentration reflects optimal wind conditions, supporting the role of wind as a complementary off-grid option.

5 Discussion

This study provides a comprehensive GIS-based spatial and techno-economic analysis of the economic potential for the deployment of off-grid MGs in Ethiopia, evaluating three renewable technologies: mini-hydro, solar, and wind. Notably, hydro MG is identified as the most economically viable option, with LCOE values ranging from 0.09 to 0.16 \$/kWh—depending on the capacity factor (CF), as evaluated through sensitivity analyses at 25 and 46%. This

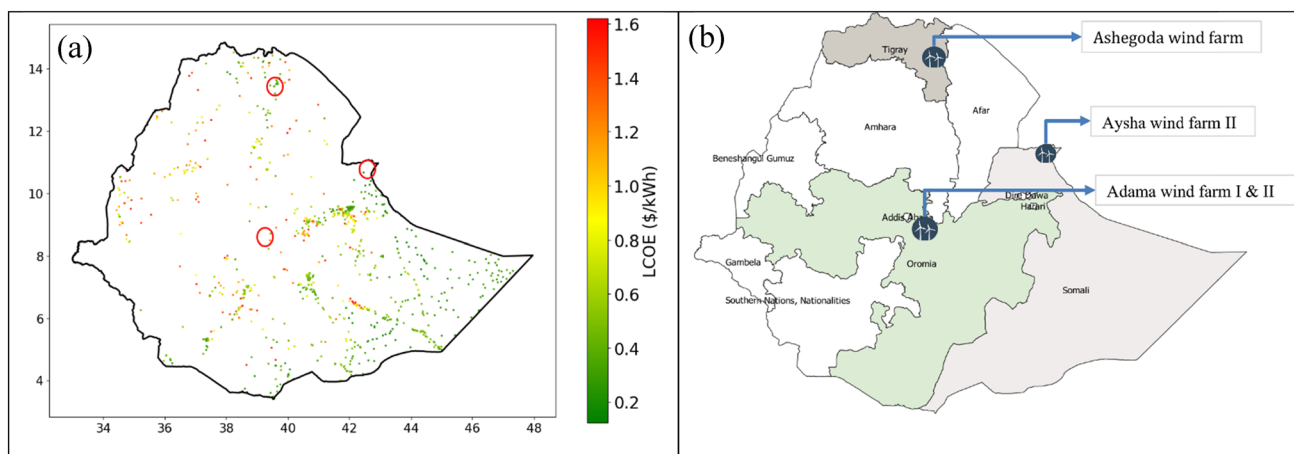


Fig. 14 Validation of MG wind sites: **a** Economically feasible MG wind sites identified in this study (left) and **b** Existing utility-scale wind farms in Ethiopia (right [37, 77])

range, reflecting hydrological variability, offers a statistically robust confidence interval, surpassing the granularity of prior assessments. The LCOE for hydro MG in Ethiopia demonstrates strong economic potential, particularly when compared to grid extension costs in remote areas, which often exceed 0.50 \$/kWh [25]. The MG hydro LCOE values (0.09–0.16 \$/kWh) aligns with IRENA's range for small hydro projects in developing countries (0.02–0.10 \$/kWh), though higher than pico-hydro (up to 0.27 \$/kWh or more) [49].

Solar MGs, predominantly viable in northern and eastern regions, yield an LCOE of 0.15–0.22 \$/kWh (mean 0.17 \$/kWh), resonating with global ranges (0.06–0.26 \$/kWh, ESMAP [63]) and regional studies (e.g., 0.30–0.65 \$/kWh in Tanzania [78], \$0.63/kWh in Kenya [74], 0.147 \$/kWh in Morocco [10]). The study's findings also correlate with the ESMAP report, which shows a reduction in solar hybrid MG costs from \$0.55 to 0.38 \$/kWh between 2018 and 2021, with further reductions expected.

Wind MGs, however, have a more spatially constrained potential compared to hydro and solar MGs. The eastern and northern regions of the country are particularly well-suited for wind MG development due to their favorable wind conditions as shown in Fig. 2b. Nevertheless, the LCOE for wind MGs in the selected economically feasible sites in Ethiopia shows significant variability, ranging from 0.12 to 1.75 \$/kWh, with an average LCOE of 0.69 \$/kWh. The wide range in LCOE is attributed to the spatial variability of wind speeds across the country.

However, the LCOE estimates for wind MGs are in agreement with the findings of Johannsen et al. [74], who reported an LCOE of 1.14 \$/kWh for wind MGs in Kenya. The estimates in this study also align with the lower end of the LCOE range reported by Mentis et al. [67], who estimated that the LCOE for wind MGs varies between 0.04 and 0.17 \$/kWh across the African continent. The higher LCOE for wind MG compared to hydro and solar MG can be explained by the low wind capacity factor due to the intermittency of wind energy, and also by the higher initial capital cost due to the relatively expensive wind turbines, as discussed in Sect. 3.3.1. Table 5 presents comparative data on the LCOE of MG technologies in different regions, including Ethiopia and other countries with similar energy contexts. This comparison highlights Ethiopia's competitive position in the MGs, with costs comparable to those in other developing countries.

A key strength of this study is its multi-criteria GIS framework, which integrates technical (resource potential), social (population density), and environmental (land cover, slope) factors with unprecedented resolution. Unlike prior studies that have focused solely on technical feasibility[21] or cost comparisons [26], this study provides a nationwide perspective that is currently lacking in the field of MG site selection.

Despite its strengths, this study has limitations. The results are dependent on the accuracy of the input datasets. Given the inherent uncertainties in remote sensing data, energy yield calculations, and population estimates may be subject to some error. Additionally, the study does not incorporate electricity demand and the ability to pay, which are crucial for the long-term success and sustainability of MG projects.

Nonetheless, the findings have important implications for Ethiopia's energy policy and rural electrification strategies. By identifying high-potential areas for MG deployment, the study provides a roadmap for targeted investments and policy interventions. Moreover, integrating these findings into national electrification planning can enhance the effectiveness of least-cost electrification strategies, ultimately supporting Ethiopia's efforts to achieve universal electricity access [2].

Table 5 Comparative data on MG cost competitiveness in similar regions

Country	MG technology	LCOE (\$/kWh)	Source
Global	Solar	0.06–0.26	ESMAP [63]
The continent of Africa	Wind	0.04–0.17	Mentis et al. [67]
Africa and Asia	Solar hybrid	0.38	ESMAP [15]
Ethiopia	PV/Battery System	0.25	Beza et al. [18]
Nigeria	Hydro, solar and wind combined MGs	0.31–0.46	SEforALL [79]
Kenya	Solar PV	0.63	Johannsen et al. [74]
Kenya	Wind	1.14	Johannsen et al. [74]
Tanzania	Solar PV	0.30–0.65	Moner-Girona et al. [78]
Morocco	Hydro, solar and wind hybrid MG	0.147	Allouhi et al. [10]
India	Solar PV	0.3	Adamopoulou et al. [80]

6 Conclusion

This study developed and applied a robust GIS-based framework to identify economically feasible off-grid MGs in Ethiopia, evaluating mini-hydro, solar, and wind technologies. By integrating high-resolution spatial data with technical, social, and environmental factors, the proposed methodology offers an innovative approach that goes beyond traditional site-selection methods. The LCOE analysis was used to evaluate the economic viability of selected MG sites.

The findings show that hydro MG is the most cost-effective renewable MG deployment, with an LCOE of 0.09–0.16 \$/kWh, harnessing Ethiopia's hydro resource to electrify 5 million people. Solar MG, with a mean LCOE of 0.17 \$/kWh (0.15–0.22 \$/kWh), is also viable, particularly in the northern and eastern regions, and can serve 7.2 million people. Wind energy, while more spatially constrained, remains economically viable in certain high-wind regions, with LCOE values ranging from 0.12 to 1.75 \$/kWh, providing electricity to about 4.8 million people.

Key spatial factors—population density, land cover suitability, terrain slope, infrastructure proximity, and grid expansion trajectories—anchor this analysis, ensuring MGs target high energy demand, minimal environmental constraints, and accessibility to minimize installation and operating costs. By prioritizing sites beyond the grid's near-term reach, the study ensures MG sustainability, aligning with NEP 2.0. For policymakers, the prescription is clear: prioritize hydro across river-rich landscapes, deploy solar MG in northern/eastern regions, and harness wind as a strategic complement. Integrating these findings into NEP 2.0 via least-cost geospatial targeting can improve rural electrification, by directing investments to optimal locations.

This research significantly enhances the existing methodologies for site selection in the field of off-grid electrification, presenting a comprehensive and detailed approach that can be applied on a nationwide scale. While the findings are tailored to the context of Ethiopia, the methodology serves as a scalable blueprint for SSA, where there is a similar abundance of resources alongside significant electrification deficits. It supports policy and investment decisions by offering a systematic, data-driven approach that can enhance efforts toward achieving universal electricity access, particularly in rural and remote areas where grid extension is not feasible.

Author contributions Adugnaw Lake Temesgen (ALT) contributed with Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Visualization, Data curation, Writing—original draft, Writing—review & editing. Getachew Bekele (GB) contributed with Supervision, Writing—review & editing.

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Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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